

MIS: Error-Driven Difficulty Scoring for Agricultural Object Detection

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Abstract—We propose MinImageScorer (MIS), an error-driven framework that scores per-object and image difficulty by combining model confidence and localization penalties on training-set failure signals. Validated on MinneApple and Global Wheat Head (GWHD), MIS reliably ranks objects and images by difficulty and correctly identifies domain- and size-level failure patterns without requiring any additional information. When used to guide copy-paste augmentation, MIS-selected objects outperform random selection on MinneApple ($\Delta AP = +0.013$); however, both strategies fall below baseline. On GWHD, domain shift renders copy-paste ineffective regardless of selection strategy. These findings establish MIS as a reliable diagnostic for structural difficulty in agricultural detection, while also revealing the conditions under which copy-paste augmentation is unlikely to help.

Index Terms—object detection, copy-paste augmentation, difficulty scoring, agricultural detection, small objects, domain shift

I. INTRODUCTION

Agricultural detection benchmarks present structural challenges absent from general-purpose datasets. MinneApple [1] is dominated by densely packed sub-resolution apple instances where failure is driven primarily by scale. GWHD [2] aggregates images from seven geographic domains, causing systematic performance drops on minority-domain images. Copy-paste augmentation yields gains on MS COCO [3], [4], but applied blindly to structurally constrained datasets it risks compounding failure modes: pasting sub-resolution objects amplifies scale noise; pasting cross-domain objects amplifies distributional mismatch.

We address this gap with two contributions. First, we introduce **MinImageScorer** (MIS), a per-object difficulty score derived directly from model failure signals. We validate that MIS correctly identify dataset structural difficulty without access to any additional information. Second, we conduct a controlled study showing that score-guided copy-paste outperforms random selection on the MinneApple dataset, while both strategies degrade below baseline. On the domain-shifted GWHD dataset, copy-paste is ineffective regardless of selection strategy. Taken together, these results show that MIS correctly identifies where a model fails, and that the failure mode of a dataset itself—not the selection strategy—determines whether copy-paste can succeed.

II. METHODOLOGY

A. Scoring Formulation

Let $G(I)$ be the set of ground-truth objects in image I and $\mathcal{P}(g) = \{p : \text{IoU}(p, g) \geq \tau\}$ the set of predictions matched

to ground-truth object g . Denoting $c_p = \text{conf}(p)$ and $u_p = \text{IoU}(p, g)$, the *Object Difficulty Score* is defined as:

$$S_{\text{obj}}(g) = \begin{cases} \min_{p \in \mathcal{P}(g)} [\alpha(1-c_p) + \beta(1-u_p)], & \mathcal{P}(g) \neq \emptyset, \\ \alpha + \beta, & \text{otherwise,} \end{cases} \quad (1)$$

where α and β weight the confidence and localization penalties, respectively. Missed objects (no prediction with $\text{IoU} \geq \tau$) receive the maximum score $\alpha + \beta$. The *Image Difficulty Score* combines mean object difficulty with the false-positive rate:

$$S_{\text{img}}(I) = w_1 \cdot \frac{1}{|G(I)|} \sum_{g \in G(I)} S_{\text{obj}}(g) + w_2 \cdot \text{FPrate}(I). \quad (2)$$

Object scoring uses a low confidence threshold ($\tau = 0.01$) to capture near-miss predictions; FP rate is computed at the optimal operating threshold to reflect realistic deployment errors. Weights α, β, w_1, w_2 are calibrated by maximizing Pearson correlation between S_{img} and per-image miss rate and FP rate on the training set, requiring no domain-specific prior knowledge.

B. Score-Guided Copy-Paste Pipeline

A baseline YOLOv8s detector is first trained on the target dataset. Inference on the full training set yields per-prediction confidence and IoU values, from which S_{obj} and S_{img} are computed for every annotated object and image.

Following the same pipeline as the author of [3] have done we will extend the dataset size by using Copy and Paste pipeline. The special of our implementation is that instead of choosing a number of random objects in a random image and copy-paste it multiple times in random locations with the same image, we will use the score to change the probability of selecting an object and a image to copy and paste.

Under our *score-guided copy-paste*, the probability to choose an object and a background image are sampled proportionally to their difficulty score via exponential weighting—objects with high S_{obj} are drawn more frequently, deliberately concentrating augmentation effort on instances the baseline model most consistently fails to detect or localize. This new data then used to retrain a fresh YOLOv8s model under identical hyperparameters, ensuring that any performance difference is attributable to selection strategy alone.

III. SCORE VALIDATION

A. Correlation with Detection Failures

We can see in Table I that all correlations are strongly positive, confirming that the score reliably captures image-level difficulty.

TABLE I
PEARSON CORRELATION OF S_{img} WITH ERROR RATES

Dataset	$r(\text{score, miss})$	$r(\text{score, FP})$
MinneApple	0.698	0.688
GWHD	0.816	0.818

B. Domain and Size-Level Validity

Domain-level (GWHD). Part B Table II show that the harder domains (lower AP) have higher Mean S_{img} . This indicate a strong relation between our scoring with GWHD failed mode. **Size-level (MinneApple).** Part A Table II shows that S_{obj} is higher for small objects than medium objects, which mean our scoring has correctly matched the known ap between these size categories.

Together, these results demonstrate that MIS can identify the primary structural difficulty at both domain and object-size granularity—using only training-set model failure signals.

TABLE II
VALIDATION OF SCORING METRICS ACROSS SCALES AND DOMAINS

Part A: MinneApple Size Analysis (S_{obj})							
Metric	Small ($\leq 32\text{px}$)				Medium ($\leq 96\text{px}$)		
AP Score	0.300				0.592		
Obj. Diff. Score (S_{obj})	0.378				0.222		
Part B: GWHD Domain Analysis (Mean S_{img} vs. AP)							
Domain	rres_1	ethz_1	inrae_1	arv_3	usask_1	arv_1	arv_2
AP	0.572	0.550	0.523	0.466	0.408	0.403	0.365
Mean S_{img}	0.200	0.227	0.251	0.294	0.312	0.331	0.424

IV. AUGMENTATION EXPERIMENTS

A. Setup

We use YOLOv8s initialized from COCO pretrained weights. We will use copy and paste augmentation to increase the dataset size by 30%. For each new image, 2–8 new objects are chosen to be pasted, applied Scale jitter, random flipping following Ghiasi et al. [4]. The only difference between the two experimental conditions is whether source objects are sampled uniformly (*Random CP*) or proportionally to S_{obj} (*Score CP*).

B. Results

MinneApple. Table III show that Score-guided copy-paste outperforms random selection across sub-metrics, confirming that difficulty-weighted sampling directs augmentation toward higher-signal objects. However, both conditions fall below baseline indicates degraded localization precision from pasting objects without boundary blending. The AP_S drop we believe reflects that score-selected objects are predominantly

TABLE III
COPY-PASTE AUGMENTATION RESULTS

Dataset	Condition	AP	AP_{50}	AP_{75}	AP_S	AP_M
MinneApple	Baseline	0.439	0.771	0.452	0.300	0.592
	Random CP	0.415	0.741	0.427	0.279	0.563
	Score CP	0.428	0.759	0.440	0.286	0.581
GWHD	Baseline	0.507	0.922	0.491	0.136	0.501
	Random CP	0.486	0.905	0.456	0.118	0.480
	Score CP	0.484	0.905	0.455	0.098	0.477

sub-resolution and cannot be made more learnable through repetition alone—a constraint imposed by the dataset itself, not by the selection strategy.

GWHD. Score-guided and random conditions are statistically indistinguishable and both fall below baseline. Score-guided sampling concentrates objects from domain-minority sources, amplifying existing domain imbalance rather than correcting it. Feature-space analysis confirms that GWHD augmented images suffer great distributional displacement, consistent with the larger performance gap. This we believe indicated that a domain-shifted dataset, selection quality is irrelevant if the fundamental problem is covariate shift between augmented and test images.

V. CONCLUSION

MinImageScorer reliably quantifies detection difficulty from model failure signals, validated across image, domain, and object-size-level difficulty. Score-guided copy-paste outperforms random selection on MinneApple, confirming difficulty-weighted sampling improves selection quality. Nevertheless, both strategies degrade below baseline on both datasets: the dominant failure modes—sub-resolution scale limits on MinneApple and cross-domain shift on GWHD—are structural and cannot be addressed by pasting objects whose augmented images are already displaced from the test distribution.

The practical implication is direct: apply copy-paste only when high-score objects are above the resolution floor and within the test distribution. MIS provides this diagnostic before augmentation is applied.

Future work should apply MIS scores to guide generative augmentation methods—such as diffusion-based synthesis or GAN-conditioned inpainting—to produce hard-object instances that remain within the test distribution, directly addressing the distributional shift that limits copy-paste on these benchmarks.

REFERENCES

- [1] N. Hani, P. Roy, and B. Bhanu, “MinneApple: A benchmark dataset for apple detection and segmentation,” *IEEE Robotics Autom. Lett.*, vol. 5, no. 2, pp. 852–858, Apr. 2020.
- [2] E. David et al., “Global Wheat Head Detection (GWHD) dataset,” *Plant Phenomics*, vol. 2021, Art. no. 9848507, 2021.
- [3] M. Kisantal, Z. Wojna, J. Murawski, J. Naruniec, and K. Kyriakopoulos, “Augmentation for small object detection,” in *Proc. Int. Conf. Adv. Comput., Commun. Informat. (ICACCI)*, 2019, pp. 193–197.
- [4] G. Ghiasi et al., “Simple copy-paste is a strong data augmentation method for instance segmentation,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2021, pp. 2918–2928.